MLN601Madipoju PrathyushaBrief1

0.1 Business Understanding

The major goal of this report is to predict the red wine quality depending on the physiochemical properties. It has been evaluated on the basis of scale and its objective is to develop a predictive algorithm that evaluates the quality of wine using some important features. ### Problem Statement The wine organization tends to be competitive in determining wine quality. This evaluation could improve their production and other marketing demands. In the traditional method, the quality of wine is assessed by humans depending on smell, taste, and other important characteristics. Now this method has been changed by implementing machine learning algorithms to determine the wine quality depending on the important features (Dahal, Dahal, & Banjade, 2021). In this project, the quality is predicted based on 11 physiochemical characteristics like alcohol content, sugar content, acidity, and sulfur dioxide levels. By understanding these characteristics wine manufacturers could able to optimize and monitor their production process resulting in delivering good quality products (Karekkat, 2023). ### Objectives • To build a machine learning model that is Linear Regression for predicting the red wine quality. • Identify the features that influence better on wine quality (Gupta, 2023). • Evaluate the reliability and accuracy of the LR model with metrics like R-squared and Mean Squared Error.

0.2 Data Understanding

Load Data

[1]:
```
# @title Default title text
# Import required libraries
import pandas as pd

# Load the wine data dataset
wine_data = pd read_csv( winequality-red.csv , sep= ; )

# Display the first few rows of the dataset
wine_data head()
```

[1]:
```
fixed acidity volatile acidity citric acid residual sugar chlorides \
                 0     7.4   0.70  0.00  1.9   0.076
                 1     7.8   0.88  0.00  2.6   0.098
        2     7.8 0.76 0.04 2.3 0.092 3 11.2 0.28 0.56 1.9 0.075
 4           7.4            0.70        0.00           1.9     0.076
 free sulfur dioxide total sulfur dioxide density pH sulphates \
0               11.0 34.0 0.9978 3.51 0.56
1               25.0 67.0 0.9968 3.20 0.68
2               15.0 54.0 0.9970 3.26 0.65
3               17.0 60.0 0.9980 3.16 0.58
```

```
4                11.0 34.0 0.9978 3.51 0.56
   alcohol quality
0      9.4 5 1

9.8 5 2 9.8

5 3 9.8 6

4     9.4   5
```

Pandas library has been imported for data analysis and manipulation. It is important for reading, exploring and cleaning datasets. wine quality dataset is the file read using pd.read_csv(). wine_data.head() displays the first few rows of the dataset.
Describe Data

[2]:
```
# Display the data type dataset
print("\nData Type Of Wine Data: ")
print("======================")
wine_data info()


# Display the shape of the dataset
print("\nShape of the dataset: ")
print("=======================")
print(wine_data shape)

# Summary Statistics for dataset
print("\nSummary Statistics: )
print("====================")
wine_data describe()
```

```
Data Type Of Wine Data:
======================
<class
'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 12 columns):
# Column                Non-Null Count Dtype
--- ------              -------------- -----
0 fixed acidity         1599 non-nullfloat64
1 volatile acidity      1599 non-nullfloat64
2 citric acid           1599 non-nullfloat64
3 residual sugar        1599 non-nullfloat64
4 chlorides             1599 non-nullfloat64
5     free sulfur dioxide 1599 non-null float64
6     total sulfur dioxide 1599 non-null float64
7 density               1599 non-null float64
8 pH                    1599 non-null float64
```

```
9 sulphates            1599 non-null float64
10 alcohol             1599 non-null float64
11 quality             1599 non-null int64
dtypes: float64(11), int64(1)
memory usage: 150.0 KB Shape
of the dataset:
=======================
(1599, 12)

Summary Statistics:
=====================
```

[2]:
```
fixed acidity volatile acidity citric acid residual sugar \
 count   1599.000000     1599.000000 1599.000000  1599.000000
 mean       8.319637       0.527821   0.270976      2.538806
 std        1.741096       0.179060   0.194801      1.409928
 min        4.600000       0.120000   0.000000      0.900000
 25%        7.100000       0.390000   0.090000      1.900000
 50%        7.900000       0.520000   0.260000      2.200000
 75%        9.200000       0.640000   0.420000      2.600000
 max       15.900000       1.580000   1.000000     15.500000
       chlorides free sulfur dioxide total sulfur dioxide density \
 count 1599.000000       1599.000000       1599.000000 1599.000000
 mean     0.087467         15.874922          46.467792   0.996747
 std      0.047065         10.460157          32.895324   0.001887
 min      0.012000          1.000000           6.000000   0.990070
 25%      0.070000          7.000000          22.000000   0.995600
 50%      0.079000         14.000000          38.000000   0.996750
 75%      0.090000         21.000000          62.000000   0.997835
 max      0.611000         72.000000         289.000000   1.003690
              pH   sulphates  alcohol       quality

 count 1599.000000 1599.000000 1599.000000
 1599.000000 mean 3.311113 0.658149 10.422983
 5.636023 std 0.154386 0.169507 1.065668 0.807569 min
 2.740000 0.330000 8.400000 3.000000 25%
 3.210000 0.550000 9.500000 5.000000
 50% 3.310000 0.620000 10.200000 6.000000 75%
         3.400000 0.730000 11.100000 6.000000 max
         4.010000 2.000000 14.900000 8.000000
```

wine_data.info() gives a dataset summary. It displays the data types and memory usage. This helps in verifying whether columns contain missing values and could able to understand the dataset structure. wine_data.shape() produces dataset dimensions. This provides the size of the dataset. wine_data.describe() develops a summary statistics for the columns present in the dataset. Details sich as mean, standard deviation, count, max, min are displayed. This is used in understanding the data spread and distribution.

0.2.1 Verify Data Quality

[3]:
```
# Handle Missing values
print("\nMissing values : )
print("==================")
print(wine_data isnull() sum())
```

```
Missing values :
==================

fixed acidity     0
volatile acidity  0 citric
acid   0 residual sugar 0
chlorides   0 free sulfur
dioxide 0 total sulfur
dioxide 0
density              0
pH                   0
sulphates            0
alcohol              0
quality              0
dtype: int64
```

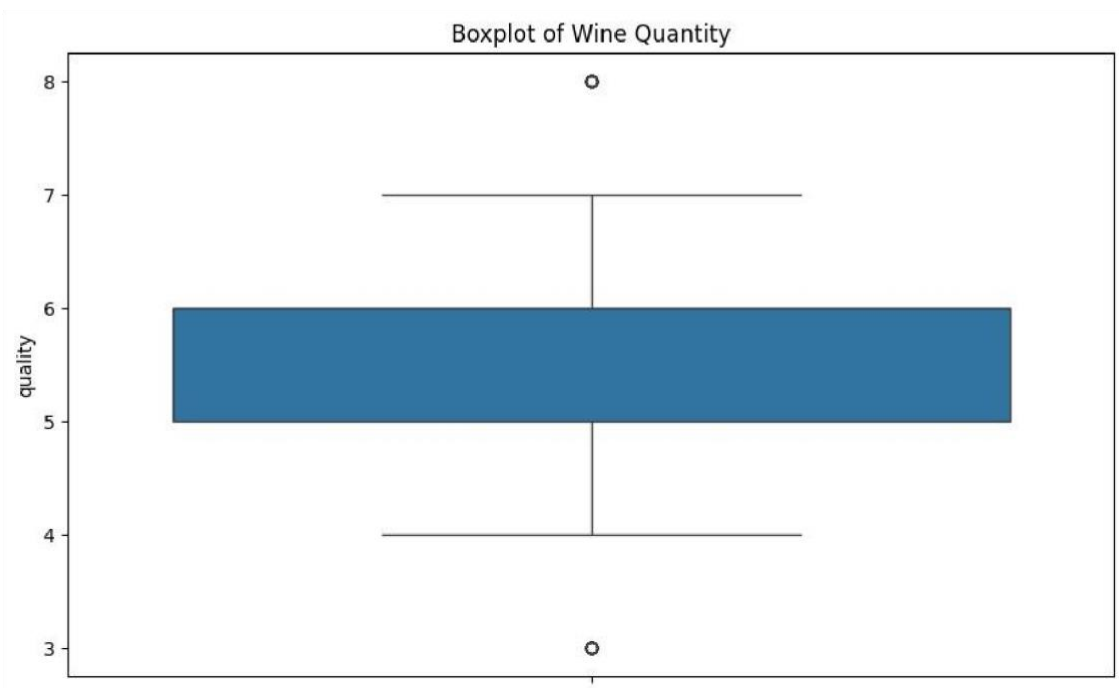
wine_data.isnull().sum() verifies missing vlaues in the dataset. It gives the missing values count. This ensures data quality as ML models are not able to handle any kind of missing data.

[4]:
```
# Handling duplicates
duplicates = wine_data duplicated() sum()
print("Number Of Duplicates: ",duplicates)


# Remove Duplicates
wine_data drop_duplicates(inplace=True)

# After removal duplicates
duplicates = wine_data duplicated() sum()
print("Number Of Duplicates after removal:  ",duplicates)
```

```
Number Of Duplicates:  240
Number Of Duplicates after removal:  0
```

duplicates = wine_data.duplicated().sum() verified the duplicate rows present in the dataset. The count of rows duplicated is returned. It aids in determining whether there is any redundant data. wine_data.drop_duplicates(inplace=True) eliminates the rows which are duplicated. duplicates = wine_data.duplicated().sum() ensures there is no remaining duplicated data. It ensures that all duplicated data has got removed successfully.

[5]:
```
# Outlier Detection
import matplotlib pyplot as plt
import seaborn as sns

# Identify outliers using the IQR method
Q1 = wine_data[ quality ] quantile(0.25)
Q3 = wine_data[ quality ] quantile(0.75)
IQR = Q3 - Q1

# Define bounds
lower_bound = Q1 - 3 * IQR
upper_bound = Q3 + 3 * IQR

# Remove outliers
wine_data = wine_data[(wine_data[ quality ] >= lower_bound) &
 ↪(wine_data[ quality ] <= upper_bound)]

# Visualizing the data after outlier removal
plt figure(figsize=(10, 6))
sns boxplot(y= quality , data=wine_data)
plt title( Boxplot of Wine Quantity')
plt show()
```

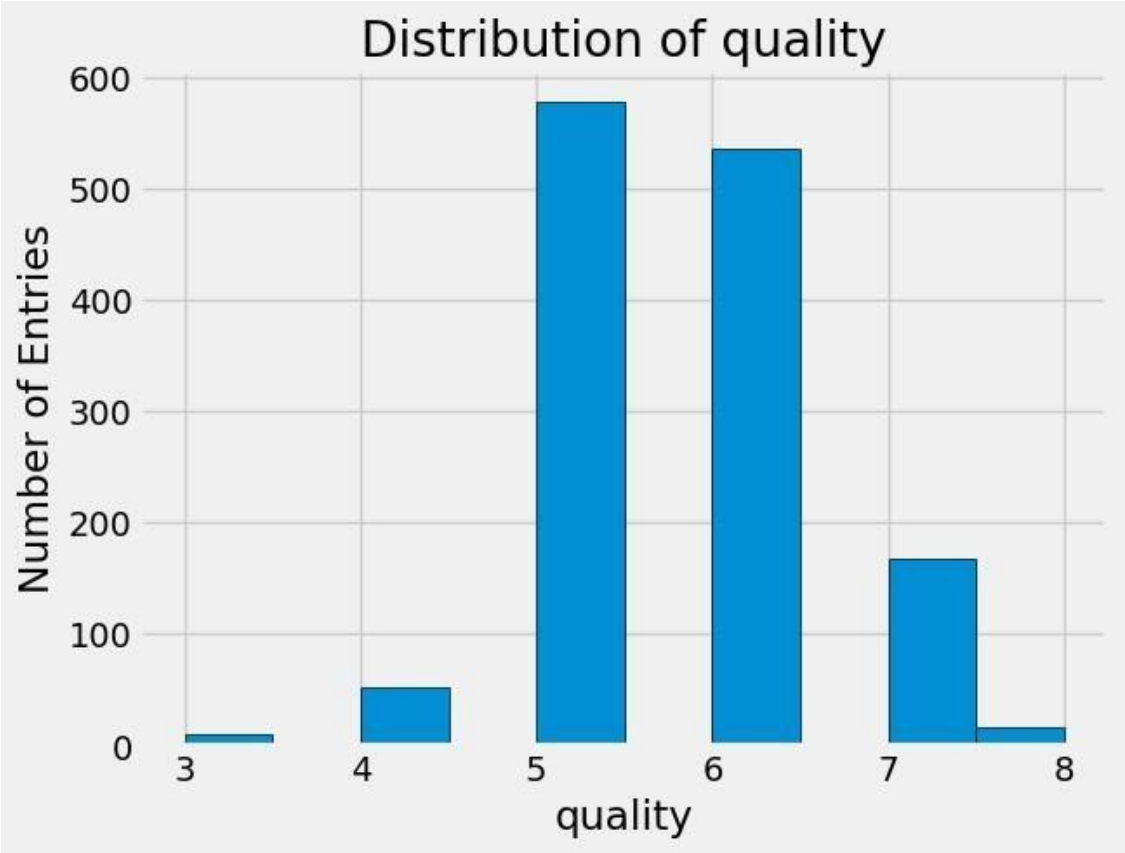


To detect outliers, IQR (Interquartile Range) is calculated. Q1 indicates the lower quartile and Q3 indicates the upper quartile. IQR is referred to be difference aming Q1 and Q3. It aids in identifying outliers. Bounds are set for identifying outliers. The formulas is used in definnig some extended range to detect outliers.

Oultiers are removed by applying the bounds and this filters the outliers that impact the performance of the model.

After this Box plot is done with the size of 10 and 6 inches. It is used in visualizing the data distribution, determining outliers. Box plot would display quartiles, pending outliers and median.

[6]:
```
import pandas as pd
import matplotlib pyplot as plt

# Count Values
def count_values_table(wine_data):
    count_val = wine_data value_counts()
    count_val_percent = 100 * wine_data value_counts() / len(wine_data)
    count_val_table = pd concat([count_val, count_val_percent round(1)], axis=1)
    count_val_table_ren_columns = count_val_table.rename(columns={0  'Count␣
 ↪Values , 1  '%  f Total Values'})
    return count_val_table_ren_columns

# Histogram
def hist_chart(wine_data, col):
    plt style.use( fivethirtyeight )
    plt hist(wine_data[col] dropna(), edgecolor= k )
    plt xlabel(col)
    plt ylabel( Number of Entries')
    plt title( Distribution of ' + col)

    # Show the plot
    plt show()

# Histogram of quality
hist_chart(wine_data, 'quality )
# Count values for quality
print(count_values_table(wine_data[ quality ]))
```

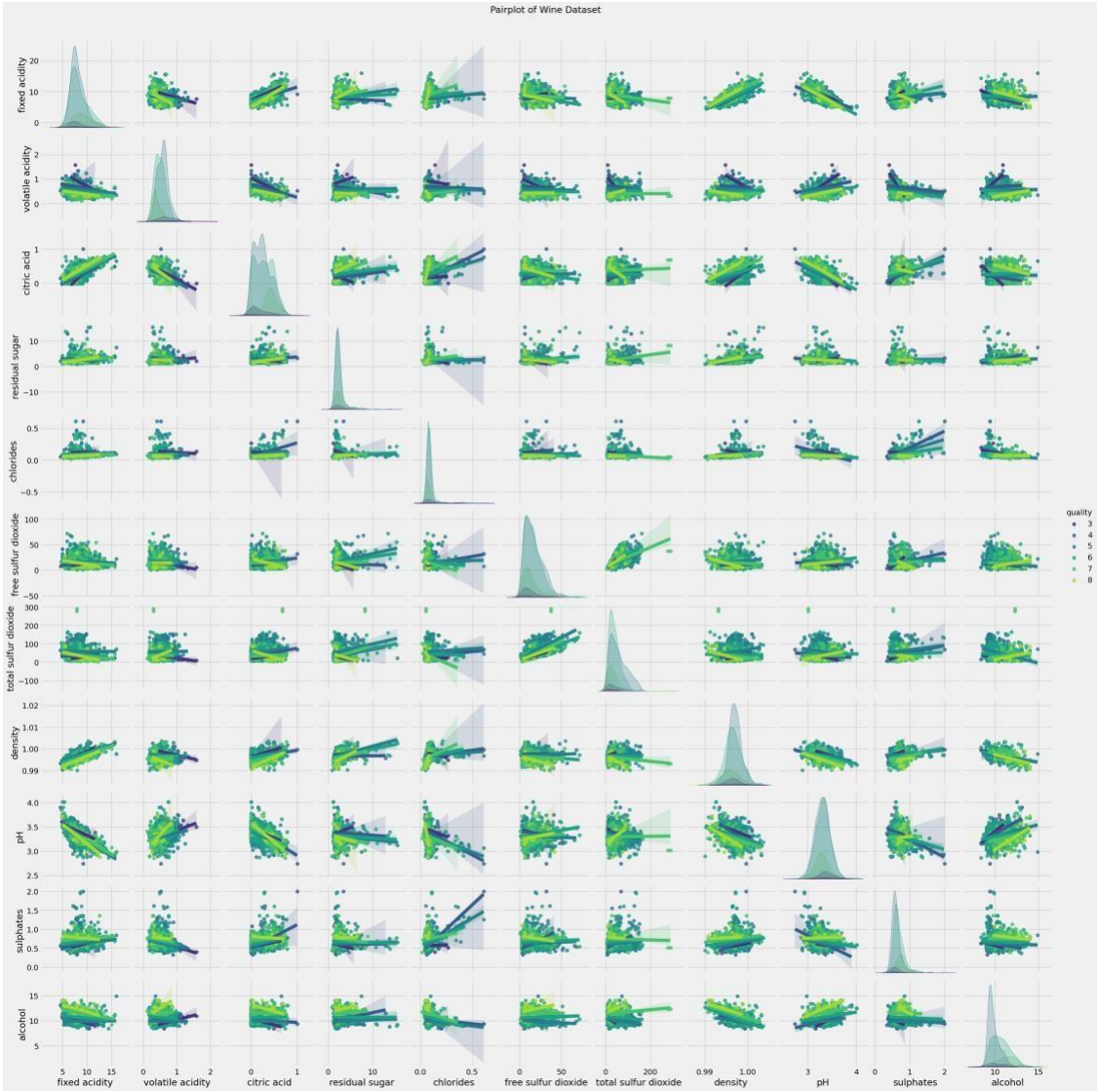



```
        count count
quality
5         577 42.5

6         535 39.4

7         167 12.3

4           53 3.9

8         17 1.3

3           10 0.7
```

wine_data.value_counts () would count the no. of occurences in the data frame. Percentage for each one of the value is calculated depending on the dataset entries. This calculates the percentage of each value depending on the entries present in the dataset.

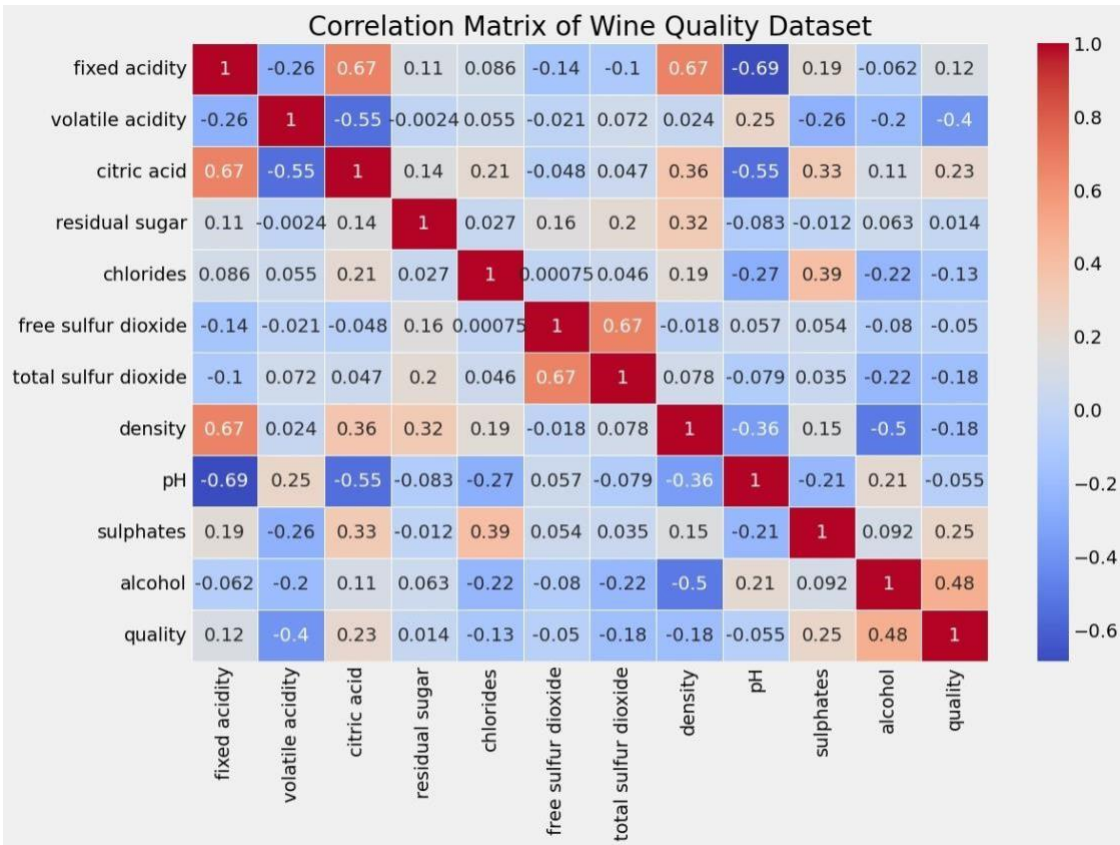
A histogram has been created to visualize the dataset distribution. X-lable and Y-label has been created. X label is for column and Y-label is for representing the bars. Title has been provided for the plot and plt. show() is used to display the plot of histogram.It prints the percentages and counts of the quality score.

```
[7]: # Relationships using pair plot sns.pairplot(wine_data.dropna(),
hue='quality', kind='reg', diag_kind='kde',␣ ↪palette='viridis')
plt suptitle( Pairplot of Wine Dataset '   =1.02)
plt show()
```



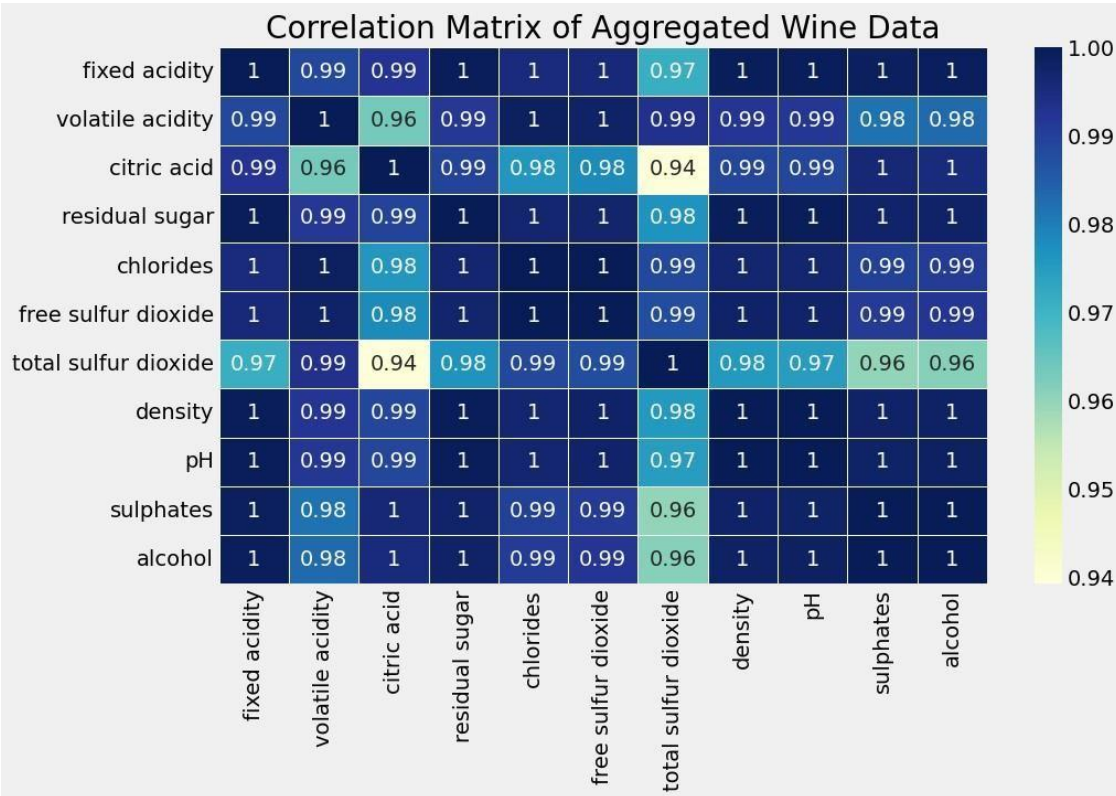
The function sns.pairplot() helps visualize the relationships among the dataset features. The plot has been colored depending on the column of quality. This permits to visualize the variations of wine quality with other different features. A regression line has been included for better visualization. Diagonal KDE plots would assist in visualizing the distribution of each one of hte variable.

```
[8]: # Correlation Matrix plt.figure(figsize=(12, 8))
correlation_matrix = wine_data.corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm',
linewidths=0.5) plt.title('Correlation Matrix of Wine Quality
Dataset')
plt show()
```



Heatmap figure has been set to have the clear visualization of correlation matrix. The figure has been set to 12 and 8 inches. wine_data.corr() used to calculate the dataset correlation matrix. Heatmap would show the relationship strength among different dataset variables. It includes the way in which it correlates with the quality of wine.

[9]:
```
# Display the aggregated correlation
matrix df_agg =
wine_data.groupby('quality').sum()

# Plot aggregated correlation matrix
plt.figure(figsize=(10, 6)) sns.heatmap(df_agg.corr(),
annot=True, cmap='YlGnBu', linewidths=0.5)
plt.title('Correlation Matrix of Aggregated Wine Data')
plt.show()
```



Dataset has been grouped by the variable quality and it computes the total of all different varaibles present in the group of quality. This aggregation is used to assess the correlations among different feautures like total acidity or alcoholic content of the wine quality and this could produce various patterns.

0.3 Data Preparation

0.3.1 Select the Data

[10]:
```
from sklearn model_selection import train_test_split

# define Feature and target
X = wine_data drop([ quality ], axis=1)
y = wine_data[ quality ]

# Split into data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size 0.2,
 ↪random_state=42)
```

The function train_test_split has been used for splitting the datset into testing and training datasets. This is important for the evaluation the ML model performance. 80% for training and 20% for testing. X axis defines the features like alcohol, acidic content etc, Y axis defines the target variable which is score of the wine quality.

0.3.2 Clean the Data

[11]:
```
from sklearn preprocessing import StandardScaler

# Standardize dataset
scaler = StandardScaler()
X_train_scaled = scaler fit_transform(X_train)
X_test_scaled = scaler transform(X_test)
```

The scaler calculates the standard Deviation and mean for all the features of the training data. Standardization is critical for ML models, particularly those that depend on distance measures like neural networks, SVM, and linear regression.

0.4 Modelling

[12]:
```
from sklearn linear_model import LinearRegression

# Initialize the model
model = LinearRegression()
```

[13]:
```
# Fit the model on the scaled training data
model fit(X_train_scaled, y_train)

# Make predictions on the test set
y_pred = model predict(X_test_scaled)
```

Linear Regression model has been imported and it is a ML model that models the relationship among the target variable and independent features. model.fit(X_train_scaled, y_train) is used for fitting the data and undergo predictions. It predicts the quality score of wine depending on the features.m

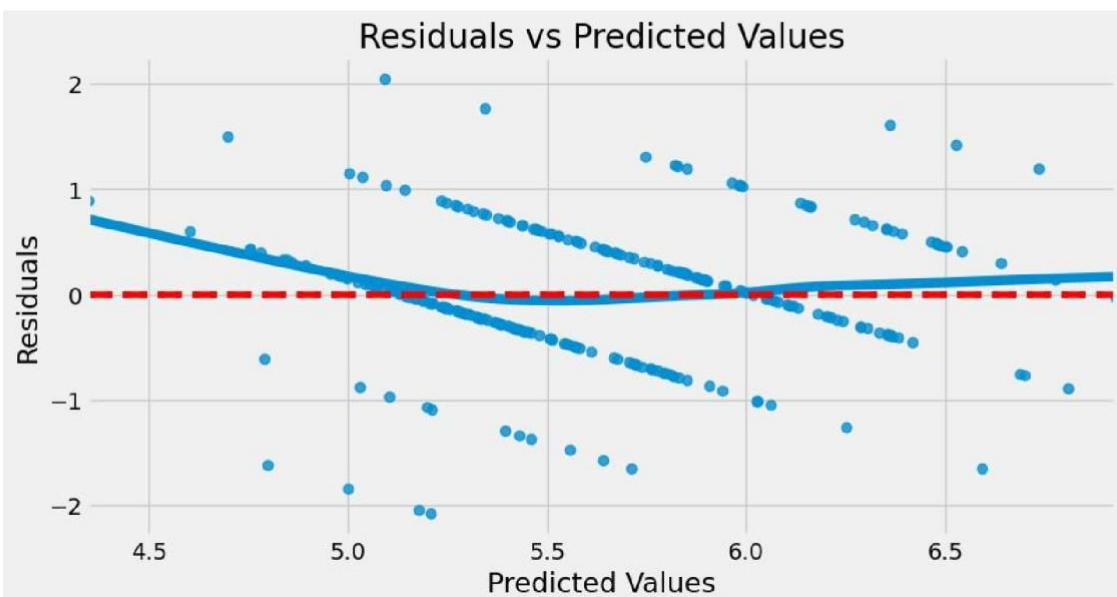
0.5 Evaluation

[14]:
```
# Evaluate the model
from sklearn metrics import mean_absolute_error, mean_squared_error, r2_score

# Evaluate the model
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

# Print evaluation metrics
print(f"Mean Absolute Error (MAE)    : {mae}")
print(f"Mean Squared Error (MSE)     : {mse}")
print(f"R-squared (R²)               : {r2}")
```

```
Mean Absolute Error (MAE)    : 0.5041409053480712
Mean Squared Error (MSE)     : 0.4310090050900971
R-squared (R²)               : 0.39153604990581903
```

Regression evaluation metrics help in evaluating the model performance. MSE (Mean Squared Error) value is 0.50, MAE(Mean Absolute Error) value is 0.431 and R (R-Squared) value is 0.39.

[15]:
```
# Plot the Residuals and Pridicted Values
plt figure(figsize=(10, 5))
sns residplot(x=y_pred, y=y_test - y_pred, lowess=True)
plt xlabel( Predicted Values )
plt ylabel( Residuals )
plt title( Residuals vs Predicted Values ')
plt axhline(y=0, color= r , linestyle= -- )
plt show()
```



sns.residplot() develops a residual plot. This plot is among the predicted (x-axis) and actual values. Residuals is represented in Y-axis. Title has been added for better clairification.

[16]:
```
# Print Feature impartance coefficients =
pd.DataFrame(model.coef_, X.columns, columns=['Coefficient'])
print("Feature Importances:") print(coefficients)
```

```
Feature Importances:
                 Coefficient
fixed acidity       -0.049675
volatile acidity    -0.178494
citric acid         -0.019148
residual sugar      -0.005231
chlorides           -0.115971
free sulfur          0.048723
dioxide
total sulfur        -0.123600
dioxide
density     0.042064 pH -
0.121548
sulphates 0.148499 alcohol 0.333970
```

Model,coef_ used in extracting the weights or coefficiects assigned for each feature by LR model. The coefficients indicates the influence or significance of each one of the feature and this determines the target variable. Depending on the coefficients, the each feature's impact has predicted on the quality of wine. Alcohol content is high and this increases the prediction of wine quality. Positive impact is on free sulfur diozide, density, sulphates and alcohol. Features having negative coefficients are volatile acidity, chlorides, total sulfur dioxide, residual sugar and fixed acidity. sulphates and alcohol are the positive contributors for higher quality predictions of wine. Chlorides and volatile acidity are negative factors minimizes the predictions quality. Citric acid and residual sugar have reduced influence on the wine quality score. This interpretation aids to understand the physiochemical traits which impacts the wine quality.